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Physics Syllabus – S6-S7

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1. General Objectives

The European Schools have the two objectives of providing formal education and of encouraging pupils' personal development in a wider social and cultural context. Formal education involves the acquisition of competences (knowledge, skills and attitudes) across a range of domains. Personal development takes place in a variety of spiritual, moral, social and cultural contexts. It involves an awareness of appropriate behaviour, an understanding of the environment in which pupils live, and a development of their individual identity.

These two objectives are nurtured in the context of an enhanced awareness of the richness of European culture. Awareness and experience of a shared European life should lead pupils towards a greater respect for the traditions of each individual country and region in Europe, while developing and preserving their own national identities.

The pupils of the European Schools are future citizens of Europe and the world. As such, they need a range of competences if they are to meet the challenges of a rapidly changing world. In 2006 the European Council and European Parliament adopted a European Framework for Key Competences for Lifelong Learning. It identifies eight key competences which all individuals need for personal fulfilment and development, for active citizenship, for social inclusion and for employment:

1. Literacy competence;
2. Multilingual competence;
3. Mathematical competence and competence in science, technology and engineering;
4. Digital competence;
5. Personal, social and learning to learn competence;
6. Civic competence;
7. Entrepreneurship competence;
8. Cultural awareness and expression competence.

The European Schools' syllabi seek to develop all of these key competences in the pupils.

2. Didactical Principles

The didactical principles of the European Schools are formulated in the teaching standards of the European Schools (ref: 2012-09-D-11-en-4). For delivery the teaching standards state that the teacher:

- Uses teaching skills and creativity to inspire and motivate pupils
- Delivers well-structured lessons
- Makes an effective use of teaching time
- Employs a variety of teaching and learning methods, including technology, appropriate to the content
- Motivates pupils to be actively involved in their own learning
- Demonstrates good subject and curriculum knowledge including their national and European dimensions

The eight competences for physics are knowledge, comprehension, application, analysis, experimental work, digital competences, communication and teamwork.

To teach the competences for physics according to the teaching standards of the European Schools an inquiry-based approach to teaching and learning is strongly recommended in S6 - S7. The learning objectives listed in this syllabus, especially the competences concerning experimental work, digital and information competency, communication and teamwork cannot be achieved without a large focus on practical work.

3. Learning Objectives

Learning is not just getting more content knowledge. With learning in school, content is used to give the pupils competences to be prepared for society and work. Learning objectives for student performance therefore arise out of three dimensions: the European Framework for Key Competences for Lifelong Learning outlined in section 1, the academic competences outlined in 3.1 and the Cross-cutting concepts (Interdisciplinary Connections) in 3.2. This way we hope that the pupils will become prepared to a lifelong learning.

3.1. Competences

	Competency	Key Concepts
1.	Knowledge	The student displays a comprehensive knowledge of facts
2.	Comprehension	The student displays a thorough command and use of concepts and principles in science
3.	Application	The student makes connections between different parts of the syllabus and applies concepts to a wide variety of unfamiliar situations and makes appropriate predictions
4.	Analysis	The student is capable of detailed and critical analysis and explanations of complex data
5.	Experimental work	The student can formulate hypotheses and plan and carry out investigations using a wide range of techniques while being aware of ethical issues
6.	Digital and information Competences	The student can consistently and independently find and assess the reliability of information on scientific subjects, on- and offline and can independently use appropriate software for science tasks
7.	Communication (oral and written)	The student can communicate logically and concisely using correct scientific vocabulary and demonstrates excellent presentation skills
8.	Teamwork	The student works well in a team

3.2. Cross-cutting concepts

The list of cross cutting competences places the learning objectives within a larger context which i. e. can form the basis of a cross-curricular projects. The tentative list to be taught is based on the next generation science standards in the United States (National Research Council, 2013):

	Concept	Description
1.	Patterns	Observed patterns of forms and events guide organization and classification, and they prompt questions about relationships and the factors that influence them
2.	Cause and effect	Events have causes, sometimes simple, sometimes multifaceted. Deciphering causal relationships, and the mechanisms by which they are mediated, is a major activity of science and engineering
3.	Scale, proportion and quantity	In considering phenomena, it is critical to recognize what is relevant at different size, time, and energy scales, and to recognize proportional relationships between different quantities as scales change
4.	Systems and system models	Defining the system under study—specifying its boundaries and making explicit a model of that system—provides tools for understanding the world. Often, systems can be divided into subsystems and systems can be combined into larger systems depending on the question of interest
5.	Energy and matter	Tracking energy and matter flows, into, out of, and within systems helps one understand their system's behaviour
6.	Structure and function	The way an object is shaped or structured determines many of its properties and functions
7.	Stability and change	For both designed and natural systems, conditions that affect stability and factors that control rates of change are critical elements to consider and understand

4. Content

4.1. Topics

In S6 – S7 the material is structured by topic so that relevant content and competences are acquired at depth as well as in breadth. The material from S4 and S5 is expected to be known in S6 and will only be briefly repeated at the introduction of each topic.

The students, as citizens, should later be able to discuss and decide about items which are associated with physics. Therefore, it is strongly recommended that the students undertake one project each year linked to aspects of social issues (see attainment descriptors).

It is recommended that physics and mathematics teachers coordinate their teaching order to help the students address the mathematics needed for science subjects (see assessment part).

Topic	S6	S7
Fields: Forces and Field Strength	x	
Fields: Potential, Lorentz Force, Induction		x
Interactions	x	
Oscillations and Waves: Wave equation, Huygens Principle	x	
Oscillations and Waves: Diffraction, Standing Waves		x
Quantum Physics		x

- **Fields**

An understanding of fields is foundational to modern physics. Fields exist in space, can help to explain how objects can act with a force on each other without touching, can store potential energy in a system of interacting objects, and serve as the foundation for understanding quantum physics. Many fields obey similar mathematical laws, and students can use a small set of principles about the nature of fields to interpret and explain many phenomena. These ideas about fields are developed as students study a variety of different fields. In the unit on fields, students learn about three key fields that dominate everyday life: electric, magnetic and gravitational fields. Throughout their investigations, students first learn about how fields convey forces at a distance, then how these fields can store and transfer energy.

In Years 4-5, students have explored some components of magnetic fields, such as the magnetic field from permanent magnets, solenoids, and wires. In Years 6-7, students build upon their existing ideas about magnetic fields, as well as what they have learned about the effect of force on motion, to explore electric and gravitational phenomena and explain them in terms of interactions between fields and particles. The study of fields in Year 6 begins with an exploration of the electric field and how it can be used to explain interaction between charged particles at a distance and to

store energy. Their study of electric fields begins by learning that charges can act on each other at a distance, and that this force occurs as a result of the fields that surround charged particles. After defining and calculating the electric field and the magnitude of the resulting force, students explore the role of electric fields in storing energy in electric circuits as charges move to or from a capacitor. In this context, students explore the relationship between electric potential and electric field strength. In addition to using mathematical models to relate potential to fields, students construct physical models to relate the two concepts. An application of the electric field is the electric circuit, as well as the ability to store and transfer energy via electric fields.

Throughout the unit, students should do much more than solve mathematical problems. At all stages, students should construct, analyse, and collect data in order to investigate how fields can store and transfer energy.

After they investigate electric fields, students turn to magnetic fields. Most students are familiar with magnetic fields both from previous instruction and everyday life. In the magnetic field unit, the emphasis is on the role of magnetic fields in explaining interaction at a distance (especially with respect to moving charges and current-carrying wires), and on the relationship between the direction of the magnetic field and the force on moving charges. Students should use simple vector operations like the “right hand rule” to model magnetic interactions. The second major emphasis within the unit is on the role of magnetic fields in generating electric fields, and vice versa. Students should use metaphors to understand the idea of magnetic flux and how a changing magnetic flux induces an electric field and can lead to an electric current in a wire. Circuit elements that rely on magnetic fields, such as inductors, should also be explored conceptually and represented mathematically.

Students also investigate gravitational fields by relating them to what they have learned about electric fields. Here, the emphasis is on gravitational fields as an explanatory mechanism for interaction-at-a-distance and the relationship between gravitational fields and gravitational potential energy. Students should construct mathematical and computer simulations of fields and energy to solve real-world problems.

• Interactions

Phenomena occur when objects interact, and systematically analysing these interactions gives students a powerful basis for interpreting and explaining a wide range of phenomena in everyday life. Building on their knowledge about the laws of motion, students start to use the mathematical concept of functions to build more complex models that can predict and explain the movement of objects ranging from a soccer ball on the pitch to satellites in space. Expanding on students' prior knowledge about Newton's laws and forces as vectors enables them to model and explain instantaneous changes in movement. Developing already acquired concepts like kinetic energy, potential energy, work, and power further, the concepts of force, momentum, and energy are applied on familiar scales to model interactions of objects, e.g. in collisions of a billiard balls or the bouncing of a basketball.

A remarkable accomplishment of physics is the recognition that an extraordinarily wide range of phenomena can be interpreted using the same mathematics and the same physical principles.

Thus, phenomena that involve gravitational fields and take place over the course of days or even centuries can be analysed with the same set of tools as phenomena that occur in far less than a second and involve interactions mediated by electric and magnetic fields.

Throughout this unit, students learn to apply the same set of principles to analyse a wide range of phenomena. To begin, students use the principles of kinematics and dynamics to analyse situations involving 1-D and 2-D motion. This analysis stresses the mathematics and physical concepts involved in making sense of how things move when they are subjected to forces, whether these forces arise from gravitation, contact between objects, or charged particles in electric and/or magnetic fields. Students' analytical toolbox expands when they begin to study how the interactions that occur during phenomena result in the transfer and conservation of momentum and energy between various systems and subsystems. Such conservation laws place constraints on the behaviour of interacting systems.

After learning to consistently use force and energy principles to analyse interactions, students explore the periodic motion of objects in circular motion and harmonic oscillatory motion. Like the analysis that they have conducted up until this point in the unit, the mathematical and physical principles that govern circular motion and oscillations apply across a wide range of phenomena. For this reason, students explicitly consider phenomena at the "everyday" scale, and they also consider phenomena at very large space and time scales (e.g., planetary orbits) and very small space and time scales (e.g., AC circuits).

Throughout the interactions unit, a central principle is that the same physical principles of Newton's Laws, conservation of energy, conservation of momentum and oscillations can be widely and consistently applied, and that this breadth of application of physical laws helps us to develop a coherent and consistent way to interpret a wide range of phenomena that we observe in the world around us.

Close cooperation with mathematics is intended. Teachers are encouraged to collaborate with Mathematics colleagues to use links between the subjects such as using data collected in Physics for Mathematical Modelling.

• **Oscillations and Waves**

In order to gain a decent understanding of waves, it is useful to begin with concrete examples of waves and proceed to more abstract waves such as electromagnetic waves.

The many diverse examples of waves that surround us provides students with the motivation to study wave concepts and could be a good way to begin the unit whilst reviewing wave concepts that were first introduced in S4 and S5. A first step will be to properly describe waves and distinguish between waves and oscillations. This leads to the question of how waves propagate in space which is the second subtheme. After the basics of waves are presented then wave phenomena of superposition, reflection, refraction and diffraction will be introduced. Huygens' principle should be used to help explain wave propagation and behaviour.

The wave unit has been developed as a progression from concrete mechanical waves to abstract electromagnetic waves. Introducing the photon will help students transition to the ideas of

Quantum Physics.

Previous knowledge is gained in the S4 Waves topic: Sound and Light as examples of waves with common characteristics of wavelength, frequency, speed, and amplitude (λ, f, v, A). Knowledge from math of trigonometric functions (sine and cosine) is useful.

To deepen the understanding of the student and to foster key competencies a research project can be included, surveying the different ways in which electromagnetic waves have been and are being used to transmit information. (N. Tesla, radio transmitter, television, wireless internet).

Following on from this section, the final topic on quantum mechanics will deal with matter waves, such as electrons and develop the student's appreciation of how widely wave concepts are found in our Universe.

• Quantum Physics

Quantum physics has changed the fundamentals of physics and has become the foundation of modern physics. Quantum physics explains the nature and behaviour of matter and energy at the atomic and subatomic level. The findings that have led to the formulation of quantum theory and the discussion of its interpretation provide a perfect context for demonstrating to students the fundamental relevance of physics to a technology-based society. Following the historic developments in terms of major experiments (e.g. the Franck-Hertz experiment) shows how physicists successively obtained deeper insights into the mechanisms at the atomic and subatomic levels and to demonstrate how deep the paradigm shift reached that came with these insights. Quantum physics provides a perfect context to engage students in major competencies of science such as the development and use of models, argument based on evidence, and the use of mathematical representations and computational thinking. It also allows students to address major cross-cutting concepts such as scale, proportion and quantity, systems and system models, and the nature of science and scientific enquiry.

Quantum physics is centred on the behaviour and properties of quantum objects. Quantum objects behave like both waves and particles. One prominent example of a quantum object is the photon. The photoelectric effect shows the limits of the wave theory of light in explaining all aspects of light behaviour, necessitating the assumption that light also behaves as a particle, a quantum object. In turn, electrons as particles can also show a wave-like behaviour. This builds on a prior understanding of electrons as particles and how electrons behave as a result of an interaction between a field and matter. The next step is to resolve the wave-particle duality through discussion of the stochastic nature in the behaviour of quantum objects. Using examples of historic double-slit experiments, the idea of the wave-particle duality of quantum objects as well as their major properties can be developed. These properties include: Quantization, Stochastic behaviour, Interference, Uncertainty and Non-locality. The application of quantum ideas to develop a quantum model of the atom and the use of this model to explain the structure of the periodic table mark the end of the quantum physics unit.

Quantum physics is often taught through the historical development of the subject. Although this is a feasible approach, other approaches are possible. For example, quantum objects and their properties can be introduced using examples of key experiments and their results such as double-

slit experiment(s), the Photoelectric Effect or the Compton Effect. Note that, although quantum physics may appear to require a more theoretical approach, some of the key experiments (e.g. the Photoelectric Effect and double-slit experiments) can actually be performed in the classroom, giving a more investigative approach. Students should be given the opportunity to explore these experiments, or at least work with original data from these experiments to investigate and draw conclusions. Where experiments are not available, or cannot easily be performed in the classroom, simulations can be used.

There are six different symbols in the following content part, which indicate the areas:

	Activity
	Extension
	Phenomenon
	History
	Cross-Cutting Concepts
	Digital Competence

These icons highlight different areas and are used to make the syllabus easier to read. These areas are based on the key competences mentioned in section 1 of this document

4.2. The tables

- Year S6

YEAR S6	TOPIC: Fields	Pre-knowledge: S4: circuits, current, charges, power, magnetism, field lines, electromagnetic induction, force, Newton's laws; S5: energy and work,		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
Electric field	Revise the nature of electric charge. The elementary charge $e = 1.6 \times 10^{-19} \text{ C}$	Plan and carry out investigations to determine the force between charged objects separated by a distance. Explain why this quantity of charge is described as “elementary”. Recall the quantity of elementary charge and use it in calculations.	   	Attracting/repelling charged pith balls. Electrostatic generator. Discuss the Millikan experiment and its significance for establishing the quantisation of charge. Relate the elementary charge to the idea of ions with relative charges +1, -2 etc as used in Chemistry. Discuss fundamental particles and their charges (Standard Model).
	Force between point electric charges $F = k \frac{Q_1 Q_2}{r^2}$ where $k = \frac{1}{4 \pi \epsilon_0 \epsilon_r}$	Calculate the force between two point charges.		Experiments to show the shape of electric fields visually (ex. Semolina grains or carbon powder in oil).

YEAR S6	TOPIC: Fields	Pre-knowledge: S4: circuits, current, charges, power, magnetism, field lines, electromagnetic induction, force, Newton's laws; S5: energy and work,		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
	Electric field strength $E = \frac{F}{q}$ Radial Field $E = k \frac{Q}{r^2}$ Uniform Field $E = \frac{\Delta V}{d} = \frac{U}{d}$ where d is the separation of two differently charged parallel plates.	Define electric field strength. Develop physical models for the field around a point charge and between differently charged parallel plates using field lines. Use mathematical models of electric fields to calculate the electric force between two point charges and on a point charge in a uniform field.	 	Extension to solve 3 charge problems and superpositions of fields is possible. The influence of a dielectric on the electric field could be explored.
Capacitor	The parallel plate capacitor as a charge and energy storage device $Q = C U$ $W = \frac{1}{2} C U^2$ where C is the capacitance.			Discuss lightning, Piezo-electric devices, capacitive touchscreens and other technological and natural examples of capacitance and discharge.

YEAR S6	TOPIC: Fields	Pre-knowledge: S4: circuits, current, charges, power, magnetism, field lines, electromagnetic induction, force, Newton's laws; S5: energy and work,		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
	Current and voltage as a function of time in a circuit as a capacitor is charged or discharged.	Design and construct simple circuits to investigate the factors that affect the charging time of a capacitor. Gather simple evidence to support the claim that capacitors store energy in the electric field between two conductors.	 	Demonstrate the effect of the dimensions, and the dielectric of a parallel plate capacitor on its capacitance. This can make a useful quantitative or qualitative experiment. The topic offers a good opportunity to introduce the students to the use of the Oscilloscope.
Magnetic field	Magnetic fields around moving charges The Magnetic Flux Density B The uniform field inside a solenoid $B = \mu n I, \quad n = N/l$	Conduct experiments to gather evidence that moving charges produce a magnetic field. Recall and describe the shape and directions of the magnetic field around a long straight current carrying wire and the field in and around a solenoid. Investigate by experiment the factors affecting the magnetic field inside a solenoid. Use the formula for the field inside a solenoid in calculations.	  	Oersted experiment, electron beams. Consider the similarities and differences between the fields around permanent magnets and those observed around currents. A Hall probe and a Slinky (long helical metal toy) can be used to investigate the dependence on current and coil density.

YEAR S6	TOPIC: Fields	Pre-knowledge: S4: circuits, current, charges, power, magnetism, field lines, electromagnetic induction, force, Newton's laws; S5: energy and work,		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
Gravitational field	Newton's Law of Universal Gravitation $F_G = G \frac{m_1 m_2}{r^2}$ where G is the Universal Gravitational Constant	Calculate the force between two point or homogeneous spherical masses. Use mathematical reasoning to explain how determining the magnitude of the Universal Gravitational Constant made it possible to determine the mass of the Earth knowing its dimensions.	 	Discuss the variation of the acceleration due to gravity at different positions and altitudes around the Earth. Cavendish experiment.
	Gravitational field strength is given by $g = \frac{F_G}{m}$ Radial and uniform fields. For radial fields $g = G \frac{M}{r^2}$	Define the gravitational field strength. Develop physical models for representing the field around a point mass and near the surface of the Earth in terms of field lines and discuss how the two models can be related. Use mathematical representations of gravitational fields to calculate the gravitational force between two point masses and on a mass in a uniform field.	 	Compare and contrast: The Newtonian Gravitational field differs from E and B fields in that there is no possibility of repulsion and no way to screen out the gravitational field. Discussion of how it is possible for a mass to maintain a stable orbit if it has sufficient tangential velocity such as: Solar system, satellites, comets, etc. Connection to motion along equipotentials could be explored.

YEAR S6	TOPIC: Interactions	Pre-knowledge: S4: Work, potential and kinetic energy; S5: momentum, conservation laws (energy and momentum).	
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities
Uniform and uniformly accelerated motion	$s(t) = s_0 + v_0 t + \frac{1}{2} a t^2$ $v(t) = v_0 + a t$ Independence of motion in two perpendicular directions in uniform fields Projectile motion in a uniform field	Plan and carry out investigations to determine relationships between position - time, speed - time, acceleration - time data. Analyse and interpret the data. Use mathematics to analyse changes in motion. Analyse motion in two dimensions in terms of the resultant of independent perpendicular components. Plan investigations to measure the movement of masses and charged particles in uniform fields. Analyse the data obtained. Calculate the kinematics of particles, in uniform electric or gravitational fields or both, using mathematical functions. (Calculations should not include dissipative forces.) Apply energy conservation to kinematic problems and discuss the constraints it imposes.	    Video analysis of motion including real life situations. Datalogging activities and graphical analysis. Combine with mathematics to give differentiation a nice context and give kinematics a mathematical basis. Investigate differences between the motion of particles • due to gravitational force alone, • with dissipative force, • with propulsion. Examples could include sports, golf ball trajectory, parachutes, braking distances, re-entry of space vessels, bullets, etc.

YEAR S6	TOPIC: Interactions	Pre-knowledge: S4: Work, potential and kinetic energy; S5: momentum, conservation laws (energy and momentum).		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
Uniform circular motion	Angular velocity $\omega = \frac{\Delta\theta}{\Delta t} = \frac{v}{r}$ Centripetal acceleration $a_{cent} = \frac{v^2}{r} = \omega^2 r$ Kinematics of objects in uniform circular motion - terrestrial objects - charged particles in uniform magnetic fields - orbits of masses in space	Describe and identify the properties of circular motion. Recognise circular motion in different contexts and explain as accelerated motion. Recognise that circular motion requires a centripetal resultant. Identify the centripetal force in examples of circular motion. Investigate the centripetal acceleration and force experimentally. Use mathematical representations to calculate characteristics of objects in circular motion.	     	Carousels, CD/DVD, wheels, etc. Discuss misleading ideas about the so-called “centrifugal force”. Electron beam tubes with Helmholtz Coils. Flat and banked curves. Satellites. Can be used in S7 to repeat circular motion: Aurora, Cyclotrons and other phenomena can be explored. Discuss energy conservation in horizontal and vertical circular motion.

YEAR S6	TOPIC: Oscillation and Waves	Pre-knowledge: S4: Oscillators, potential and kinetic energy, electric circuits		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
Harmonic Oscillations	Experiments to demonstrate concepts of restoring force and frequency in the simple harmonic oscillations of a pendulum and an oscillating spring $f = \frac{1}{T}, \quad \omega = 2\pi f$ Simple harmonic motion (SHM) $F = -k y,$ $a = -\omega^2 y,$ $y = A \sin(\omega t)$ where F is the restoring force, A is the amplitude of the oscillation, y is the displacement from equilibrium and k a constant.	State that oscillatory motion requires a resultant force directed towards the equilibrium position of the system at $y = 0$. Identify the forces responsible in examples of oscillatory motion. Analyse and interpret the kinematics and dynamics of an oscillating mass on a spring and a simple pendulum. Relate the SHM of an oscillating system to the projection in one dimension of uniform circular motion. Understand that for a harmonic oscillation the restoring force must be proportional to the displacement. Construct (derive) equations and solve problems related to the simple harmonic undamped oscillation.	    	Experiments such as with data loggers to show the sinusoidal motion and compare displacement, velocity and acceleration. Examples of uses of oscillators such as the pendulum clock. The motion of an undamped harmonic oscillator could be discussed in terms of the differential equation $m\ddot{y} + ky = 0$. Experimental examples of the superposition of oscillations of different frequencies could be explored such as in the behaviour of the ear drum, etc. Non-harmonic oscillations could be considered.

YEAR S6	TOPIC: Oscillation and Waves	Pre-knowledge: S4: Oscillators, potential and kinetic energy, electric circuits		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
	Energy stored in the simple harmonic oscillations of mechanical systems The role of an external periodic force in transferring energy to an oscillating system Resonance occurs when a natural frequency of the system is the same as that of the driving force	Identify and describe energy conservation in oscillating systems. Understand resonance as the point at which energy transfer between the forcing oscillation and the system is most efficient. Define the condition for resonance in terms of forcing frequency and natural frequency.	   	In experiments examine energy conservation in undamped oscillating systems. Energy dissipation from damped systems could be considered. Tuning forks to illustrate natural frequency Singing glass, radio receivers and emitters etc Design engineering of structures to resist earthquakes, traffic, wind exposure, ...

YEAR S6	TOPIC: Oscillation and Waves	Pre-knowledge: S4: Oscillators, potential and kinetic energy, electric circuits		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
	Electromagnetic energy exchange between the electric field of the capacitor and the magnetic field of the inductor in an LC circuit Energy in the magnetic field $W = \frac{1}{2} L I^2$ where L is the inductance. (Recall energy in the electric field $W = \frac{1}{2} C U^2$ where C is the capacitance.)	Collect and analyse data from LC circuits. Compare the behaviour of the LC circuit to a mass oscillating on a spring. Explain the behaviour of the current and voltage in an LC circuit.	  	Students build circuits and use an oscilloscope. Dataloggers allow students to produce mathematical models of these systems from experiment data. Simulations could be helpful to encourage students to identify patterns and link concepts across different topics.
	Electromagnetic oscillations can be sustained using a driving AC supply continually inputting energy in LCR circuits			
	Resonance and the natural frequency of LC oscillator $\omega_0 = \frac{1}{\sqrt{LC}}$	Use a conceptual model of a driven series RLC circuit to relate the driving frequency to the response of the electric current oscillations in the circuit. (Content limit: students should use the analogue to a mechanical oscillator to arrive at the natural frequency of the LC circuit. Other characteristics, such as impedance, are not investigated).		

YEAR S6	TOPIC: Oscillation and Waves	Pre-knowledge: S4: Oscillators, potential and kinetic energy, electric circuits		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
Description of waves	Wave as spreading of an oscillation in space with the following characteristics: period T, frequency $f = 1/T$, wavelength λ, amplitude A, displacement $y(x, t)$, phase φ, speed of propagation of a wave $v = f \lambda$ Longitudinal and transversal waves	Identify waves in the world around. Find and describe similarities and differences in the behaviour and characteristics of waves. Explain how energy is transported by a progressive wave without the bulk movement of mass in a system of coupled oscillators. Use a model of waves to explain the difference between a transverse and a longitudinal wave.	   	Ropes, ripple tanks, loudspeakers, and so on are useful to illustrate and clarify. Computer simulation can also be used. Measurements of speed of sound. Long spring to illustrate transverse versus longitudinal waves. Context: earthquake with primary and secondary waves.
Sinusoidal Wave Equation	Sinusoidal waves in time $y(t)$ and space $y(x)$ and mathematical expression $y(x, t) = A \sin \left[2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right) \right]$	Construct and interpret graphical representations of a sinusoidal wave so as to distinguish between the temporal, $y(t)$, and spatial, $y(x)$, descriptions of the wave. Distinguish between the velocity of propagation and the time dependent oscillation velocity of one of the coupled oscillators.	 	Use of differentiation to find the speed and acceleration of an oscillator. Computer simulations and dynamic geometry apps such as GeoGebra are useful to make the maths more concrete for students. Students can type in the wave function and observe what happens in y-t, and y-x space.

YEAR S6	TOPIC: Oscillation and Waves	Pre-knowledge: S4: Oscillators, potential and kinetic energy, electric circuits		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
Doppler Effect	Doppler effect of Mechanical waves $f_{observer} = f_{source} \left(\frac{v \pm u_{obs}}{v} \right)$ $f_{observer} = f_{source} \left(\frac{v}{v \pm u_{source}} \right)$ Doppler effect of Electromagnetic Waves	Use mathematical and physical arguments to derive the Doppler Effect formulas and use them. Identify examples of the Doppler Effect in nature and technology and explain them.		Examples for Doppler effect: ambulance, Doppler imaging, use by bats to hunt, rotation of the Sun, Doppler speed traps.

YEAR S6	TOPIC: Oscillation and Waves	Pre-knowledge: S4: Oscillators, potential and kinetic energy, electric circuits		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
Behaviour of waves	Superposition of waves	Identify examples of superposition of waves.		Superposition could be illustrated with a wave-machine, long ropes, loudspeakers (beats and interference). Oscilloscopes or dataloggers could be used to analyse the component frequencies in sounds.
	Huygens' principle	Use geometrical representations to show wave properties such as reflection and refraction.		Use ripple tank with wave generators to illustrate reflection, refraction, diffraction, superposition. Fourier analysis and synthesis can be presented in this context.
	Reflection $\alpha_{incidence} = \alpha_{reflection}$	Solve simple quantitative problems involving reflection and refraction of light.		
	Refraction $\frac{\sin \alpha_1}{\sin \alpha_2} = \frac{n_2}{n_1} = \frac{c_1}{c_2}$	Identify examples of reflection and refraction of waves and explain them in terms of changes in direction and/or speed.	 	Snell-Descartes' Law, critical angle and Total Internal Reflection examples and experiments. Probing of the Earth's internal structure by analysis of propagation of earthquake waves. Geological prospecting.
	Dispersion	Describe the dependence of refractive index on wavelength in dispersive media.		Research examples such as rainbows and prisms.

- Year S7

YEAR S7	TOPIC: Oscillations and Waves	Pre-knowledge: S4: Oscillators, potential and kinetic energy, electric circuits; S6: oscillations and waves		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
Behaviour of waves	Diffraction Young's Double Slit formula (Waves from two coherent point sources) $\frac{\Delta y}{D} = \frac{\lambda}{d}$ Grating Formula $n\lambda = d \sin \alpha_n$	Explain qualitatively the spreading of a wave around an obstacle or through a gap using Huygens' Principle and its dependence on wavelength. Explain the pattern of minima and maxima in double slit experiments. Plan, carry out and analyse the double slit experiment in order to simulate processes of scientific work and thinking. Derive the formulas in the content column and discuss the approximations and limitations of the methods used.	  	Single slit to introduce diffraction. Measuring of wavelength Use applets to help understanding. Grating effect of a DVD Spectrometer.
	Standing waves	Describe that two waves of equal frequency and amplitude moving in opposite directions superpose to form a stationary (standing) wave. Define the terms "node" and "antinode" and apply them to describe standing waves. State that the distance between neighbouring nodes (or antinodes) is $d = \lambda/2$.	 	Context for standing waves: Organ pipes and other wind and string instruments. Explore musical sounds produced by String or Wind instruments as the emergent resonances due to the physical characteristics of the instrument.

YEAR S7	TOPIC: Oscillations and Waves	Pre-knowledge: S4: Oscillators, potential and kinetic energy, electric circuits; S6: oscillations and waves																
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities															
	Specific bound cases: mechanical waves on a string and sound waves in a pipe of length L $L = n \frac{\lambda}{2}, n = 1, 2, 3, \dots$ $L = n \frac{\lambda}{4}, n = 1, 3, 5, \dots$ Attention: Care must be taken as the usage of “harmonic” and “overtone” varies between languages; “overtone” even does not exist in some languages.	Explain how a standing wave only forms - on a string of length L fixed at both ends or in a pipe with both ends closed or open if an integer multiple of $\lambda/2$ fits into L. - in a pipe where one end is open and the other closed an odd multiple of $\lambda/4$ fits into L. Solve numerical problems involving different vibrational modes of standing waves on strings or in pipes. Identify resonance modes from displacement vs. distance diagrams and draw such a diagram for a specified resonance frequency. <table><tr><th>Strings & pipes closed at both ends</th><th>Pipes open at both ends</th><th>Pipes with different ends</th></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td>---</td></tr><tr><td></td><td></td><td></td></tr><tr><td colspan="3">...</td></tr></table>	Strings & pipes closed at both ends	Pipes open at both ends	Pipes with different ends						---				...			 Experiments to show the dependence of resonant frequencies of a tensioned string or wire on the linear mass, tension and length of the string (Melde’s experiment).  For standing waves in pipes: experiments with tuning forks can be conducted and Kundt’s tube can be used to determine speed of sound waves.  IT tools present useful opportunities to explore Fourier Analysis of frequencies within a complex superposition of waves such as a musical sound.  Extensions to 2 or 3 dimensions can be explored such as cymbals, drums and other membranes.
Strings & pipes closed at both ends	Pipes open at both ends	Pipes with different ends																

...																		

YEAR S7	TOPIC: Oscillations and Waves	Pre-knowledge: S4: Oscillators, potential and kinetic energy, electric circuits; S6: oscillations and waves		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
Electromagnetic waves	Characteristics of electromagnetic waves E and B fields oscillate perpendicular to each other - Velocity of light - The waves are transverse relative to the direction of energy propagation Principal regions of the Electromagnetic spectrum: radio, microwave, infrared, visible, ultraviolet, X ray and gamma	State that an accelerating charge emits an electromagnetic wave. Describe the propagation of oscillating E and B fields from an antenna when an AC voltage is applied across the ends. Compare the interactions of electromagnetic waves in different parts of the spectrum with matter so as to classify them according to their regions. Relate different regions in the electromagnetic spectrum to different applications and natural phenomena.	    	Various examples of how an electromagnetic wave is produced by accelerating charges can be discussed. Animations can be used to illustrate Hertz-dipole. Historical experiments to measure speed of light, e.g., Rømer, Fizeau apparatus or interferometry. Fresnel mirror experiment. Antenna dipoles and Microwave ovens: examples of standing electromagnetic waves. Critical discussion of risks associated with different regions of electromagnetic radiation: which ones are real, and which are popular myths?

YEAR S7	TOPIC: Fields	Pre-knowledge: S6: Electric, magnetic and gravitational fields, field strength, forces in fields, linear and circular motion, kinematics of objects and orbits		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
Electric potential	Electric potential V at a point Potential difference is the change in potential energy of a unit positive charge moving between two points in the field. $\Delta V = \frac{\Delta E_p}{q}$	Define the electric potential at a position r as the work needed to take a unit positive charge from a reference point where $V = 0$ to the position r in the field. Describe and explain the relationship between lines of force and equipotentials.		
	Potential in the radial field of a point charge Q at a distance r from the charge $V = k \frac{Q}{r}$ Equipotentials	Determine the potential in radial field and uniform fields. Recall that V is taken to be 0 and infinite distance from Q (at $r = \infty$). Explain phenomena and calculate with this convention. Calculate the potential energy of a charge in a radial field or a uniform field. Relate work of electric force to motion across equipotentials.	 	Fluorescent light tubes under power lines. Computer simulations and experiments (such as mapping on conductive paper) to visually show equipotential lines and surfaces. Relate this concept to contour lines on maps.
	Work on charge moving in a homogeneous electric field $W = Q U$	Apply conservation of energy to electric potential.		

YEAR S7	TOPIC: Fields	Pre-knowledge: S6: Electric, magnetic and gravitational fields, field strength, forces in fields, linear and circular motion, kinematics of objects and orbits		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
	Electromagnetic induction $\varepsilon = -\frac{\Delta\Phi}{\Delta t}$ where ε is the Electromotive force (Emf). (For a solenoid, the flux through a single winding must be multiplied by the number of windings.) A coil with a changing current passing through it experiences a varying B field proportional to the current, so an Emf is induced across it. Lenz's Law	Design experiments to investigate a causal relationship between a changing magnetic field and an electric current. Explain observations in terms of electromagnetic induction and discuss energy transfer and conservation. Relate these explanations to everyday devices to transfer energy between systems. Use field line models to explain why a spinning coil of wire in an external magnetic field produces a predictable pattern of varying electric current in the wire. Carry out experiments to investigate, qualitatively and quantitatively, the voltage across a coil when a variable current passes through it. Explain observations in terms of the laws of electromagnetic induction, including Lenz's Law.	   	Examples of electromagnetic induction such as rotating coil generators, electric guitars, the transformer, metal detectors could be explored. The advantages and disadvantages of eddy (Foucault's) currents could be discussed. Examples include induction cooktops, a transformer, electromagnetic damping/braking. The physical meaning of L is not required at this point but could be explored by varying the dimensions of the coil. The similarities and differences between the role of a dielectric in capacitors and the role of iron core in an inductor are interesting to discuss.

YEAR S7	TOPIC: Fields	Pre-knowledge: S6: Electric, magnetic and gravitational fields, field strength, forces in fields, linear and circular motion, kinematics of objects and orbits		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
Gravitational field	Gravitational potential Potential difference is the change in potential energy of a unit mass moving between two points in the field $\Delta V = \frac{\Delta E_p}{m}$ Potential in a radial field at a distance r from the mass $V = -G \frac{M}{r}$ Change in potential energy in a uniform gravitational field $\Delta E_p = m g \Delta h$ Equipotentials	Define the gravitational potential at a position r in the field as the work done to take a unit mass from a reference point where $V = 0$ to a position r. Determine the potential in radial fields and uniform fields. Review motions in radial gravitational fields. Understand the concept of escape velocity. Carry out experiments to determine the energy transferred to or from a uniform gravitational field as the distance between the Earth and the object changes. Relate work of gravitational force to motion across equipotentials.	  	Discussion of “Weightlessness” in orbit. Extension to solve 3 mass problems and superposition of fields is possible. Computer simulations and other visualisations of gravitational fields and equipotential surfaces. Relate this concept to contour lines on maps.

YEAR S7	TOPIC: Quantum Physics	Pre-knowledge: S5: structure of matter; S6: Work, energy, momentum, oscillations and resonance; S7: waves, standing waves, electromagnetic spectrum.		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
Particle Behaviour of Waves	The concept of the photon. Energy of the photon $E = hf$ where h is Planck's constant The Photoelectric effect from metal surfaces Einstein's Equation $E_K = hf - W_0$ The threshold frequency f_0 Attention: An exact translation of "stopping potential" does not exist in some languages.	Use the term photon to describe the quantisation of electromagnetic energy. Apply the formula to calculate the energy, frequency and wavelength of photons. Describe the significance of the intensity and frequency of light on the photoelectrons and relate them to key experimental observations, e.g. photocurrent. Compare the predictions of a particle model with those of a wave model when applied to the observations of the Photoelectric Effect. Discuss the validity of each model. Identify E_K in Einstein's equation as maximum kinetic energy of the photoelectrons and therefore define W_0 as the work done to liberate an electron from the photo-emissive surface. Analyse data and interpret graphs that support the relationship between the kinetic energy of a photoelectron and the frequency of the light.	   	Other examples of wave quanta could be introduced such as phonons. Photoelectric Effect: Examination of the discharge of an electroscope on exposure to radiation. Other aspects and applications of the Photoelectric Effect such as the saturation current and quantum efficiency can be explored. Stopping Voltage Method: Measurement of the threshold frequency, work function and Planck constant across different metal surfaces. Computer simulations and videos are available to help students understand these concepts and simulate the classic experiment.

YEAR S7	TOPIC: Quantum Physics	Pre-knowledge: S5: structure of matter; S6: Work, energy, momentum, oscillations and resonance; S7: waves, standing waves, electromagnetic spectrum.		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
	Momentum of photons $p_{\text{photon}} = \frac{h}{\lambda}$ The Compton Effect	Interpret the observations from literature or computer simulations using conservation of momentum and energy to explain the frequency shift observed in the X-ray photons and the consequent gain of momentum by electrons. State that the results infer that photons have momentum.	 	Examples in technology and in nature can be researched such as light sails, radiation pressure in a star, inertial confinement fusion techniques and laser cooling, etc. Computer simulations are useful to make the experiment more concrete for students where access to actual experiments is not possible.
Wave behaviour of particles	de Broglie's Hypothesis "any moving particle or object has an associated wave" $\lambda = \frac{h}{p}$ The electron as an example of a particle-wave. Evidence from • Diffraction of electrons by a crystal lattice • Superposition of electrons in a double slit experiment	Extrapolate from the dual behaviour of the photon to predict the dual behaviour of a particle. Collect examples and evidence from literature, experiments or computer simulations. Compare and contrast evidence for particle and wave behaviour of matter.	  	Electron diffraction tube: Quantitative measurements could be undertaken using the equation for diffraction maxima: $2 d \sin \alpha = k \lambda$. Crystallography applications to determine structure in materials. The Transmission Electron Microscope provides an example of a real-life application of the wave behaviour of particles.

YEAR S7	TOPIC: Quantum Physics	Pre-knowledge: S5: structure of matter; S6: Work, energy, momentum, oscillations and resonance; S7: waves, standing waves, electromagnetic spectrum.		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
Wave-particle duality	Heisenberg's Uncertainty Principle $\Delta x \Delta p \geq \frac{h}{4\pi}$ $\Delta E \Delta t \geq \frac{h}{4\pi}$ Stochastic behaviour of quantum objects with particular reference to: - Interference patterns of photons and electrons - tunnel effect - the wave function description of quantum objects. Quantum objects to be considered: photons and electrons.	Discuss the consequences of the Uncertainty Principle in terms of measurement and information about the state of a system. Explain the interference pattern observed when (laser) light or electrons pass through a double slit based on the stochastic nature of quantum objects. Derive using a mathematical argument the claim that the momentum and position of a quantum object cannot both be exactly known simultaneously.		Double-slit experiments: The following three (historical) double-slit experiments allow for revisiting both photons and electrons as quantum objects: - Young's Experiment (1800): (Laser-)light of high intensity, major support for the wave theory of light; - Taylor's Experiment (1909): Light of low intensity; the original source "Interference Fringes with feeble light" can be read with high school students; - Jönsson's Experiment (1960): Interference of electrons; the experiment can be understood using computer simulations or the experimental data from Jönsson's original publication. The single slit experiment can be used as an analogy to demonstrate the relationship between variables in Heisenberg's Uncertainty Principle.



YEAR S7	TOPIC: Quantum Physics	Pre-knowledge: S5: structure of matter; S6: Work, energy, momentum, oscillations and resonance; S7: waves, standing waves, electromagnetic spectrum.		
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities	
	The implications of Quantum Nonlocality for determinism and causality The Correspondence Principle	Reason based on evidence that the measurement of a property of a quantum object has non-local implications and consequences for the classical understanding of cause-and-effect. Discuss the transition from lower numbers of quantum objects to higher numbers of quantum objects (i.e. the crossing into macroscopic physics).	  	One suitable (historic) context to discuss the implications of non-locality on our understanding of physics is the paradox formulated by Einstein, Podolski and Rosen; this context also provides the opportunity to discuss aspects of the nature of science and bridge into modern physics. Einstein-Podolski-Rosen experiment. Quantum entanglement. Teachers should research text and resources for student reference and encourage reading and learning.
Quantum Model of the Atom	Energy states of a quantum object of mass m in an infinite one-dimensional potential square well of width L $E_n = \frac{h^2}{8 m L^2} n^2$ The ground state of the system is defined as the lowest state of the system, $n = 1$. $n > 1$ states are called the excited states of the system.	Derive the equation for the energy levels of a quantum object in a one-dimensional infinite potential well.		Certain organic molecules have potentials that approximate a square well and could be mentioned to illustrate the topic.

YEAR S7	TOPIC: Quantum Physics	Pre-knowledge: S5: structure of matter; S6: Work, energy, momentum, oscillations and resonance; S7: waves, standing waves, electromagnetic spectrum.	
Subtopic	Content	Learning Objectives	Key contexts, phenomena and activities
	Realistic potentials The energy states of the hydrogen atom $E_n = -\frac{13.6 \text{ eV}}{n^2}$ Emission and absorption of light, quantization of energy, line spectra $W = hf = E_n - E_m $	Explain how the evidence of line spectra and knowledge of the Coulomb potential suggest that energy states in atoms are discrete. Analyse and Interpret data to deduce energy level structures.	 The Franck-Hertz experiment was key in confirming specifically the quantisation of energy in the atom. It is recommended that the experiment is carried out with students.  The characteristic X-ray spectra of metals provides further evidence for the quantisation of the atom.  Discuss the finite well and the classically forbidden region. These can be usefully explored through Computer Simulations.  Wherever possible, students should be provided with the opportunity to work with real data, for example on the Hydrogen spectral series.  Identify spectrometry links to Chemistry. Examine how dyes produce their colours.
	Other quantum numbers ($l, m, s, \dots$) and the Pauli Exclusion Principle	Explain the structure of the periodic table based <i>on</i> quantum numbers and the Pauli Exclusion Principle	 Relate to orbitals.

4.3. Number of Periods

The above tables for S6-S7 do not include teaching hours, because teaching time does not only depend on the content but also on the competency being taught.

The table below gives approximate times and are for guidance only. These numbers are estimated taking into account the periods dedicated to projects, school trips, etc.

Topic	Subtopic	Periods in S6	Periods in S7	Total
Fields	Electric Field	10	6	16
	Magnetic Field	8	8	16
	Gravitational Field	10	6	16
Interactions	Uniform Acceleration	20		20
	Uniform Circular Motion	12		12
Oscillations and Waves	Harmonic Oscillations	20		20
	Description of waves, Wave equation	8		8
	Doppler Effect	4		4
	Behaviour of waves	8	10	18
	Standing waves		14	14
	Electromagnetic waves		4	4
Quantum Physics	Particle behaviour of waves		14	14
	Wave behaviour of particles		8	8
	Wave-Particle Duality		8	8
	Quantum model of the atom		14	14
	Sum	100	92	192

5. Assessment

For each level there are attainment descriptors which are listed in the following table and explained by the competences. They give an idea of the level that students have to reach. They also give an idea of the kind of assessments that can be done. Pupils should be assessed in a broad variety of ways throughout the year, to give a wide-ranging picture of each pupil's attainments, strengths, and areas for further work.

Assessment is summative when it is used to evaluate student learning at the end of the instructional process or of a period of learning. The purpose is to summarise the students' achievements and to determine whether, and to what degree, the students have demonstrated understanding of that learning. Summative assessment evaluates the student's learning by the end of term exams.

Assessment is formative when either formal or informal procedures are used to gather evidence of learning during the learning process and are used to adapt teaching to meet student needs. The process permits teachers and students to collect information about student progress and to suggest adjustments to the teacher's approach to instruction and the student's approach to learning.

Formative assessment takes place in almost every lesson of the school year and should include the following:

- Lab reports
- Presentations
- Tests of subject content
- Tests of practical skills
- Self and peer evaluation

The competences are expressed in the table as a set of verbs that give an idea of what kind of assessment can be used to assess that goal. In the table with learning objectives these verbs are used and put bold, so there is a direct link between the competences and the learning objectives.

Assessing content knowledge can be done by written questions where the student has to respond on. Partly that can be done by multiple choice but competences as constructing explanations and engaging in argument as well as key competences as communication and mathematical competence need open questions or other ways of assessing.

An assignment where students have to use their factual knowledge to make an article or poster about a (broader) subject can be used to also judge the ability to critically analyse data and use concepts in unfamiliar situations and communicate logically and concisely about the subject.

Students have to be able to do an (experimental) inquiry. An (open) inquiry should be part of the assessments. Assessing designing and inquiry can be combined with other subjects.

Digital competence can be assessed by working with spreadsheets, gathering information from internet, measuring data with measuring programs and hardware, modelling theory on the computer and comparing the outcomes of a model with measured data. Do combine this with other assessments where this competence is needed.

Structure of the PHYSICS written examination paper

The total time allowed for the written papers in Physics: 3 hours (180 minutes).

The global structure of the written baccalaureate exam consists of four questions, which are subdivided into several related sub-questions:

- Q 1:** Fields (25 points),
- Q 2:** Oscillations and Waves (25 points)
- Q 3:** Quantum Physics (25 points)
- Q 4:** Mixed Topics (25 points)

The competences are grouped into the following four competence areas, with the indicated weighing:

- Knowledge and Comprehension ($\pm 36\%$)
- Application ($\pm 36\%$)
- Analysis and Evaluation ($\pm 16\%$)
- Written Communication ($\pm 12\%$).

In each area a deviation of up to 6% (1.5 marks) from the total marks for this area will be tolerated, as long as the total number of marks (25 points) is respected for each question. Not to exceed 4% (1.0 mark) should be aimed at.

Content of the Written Baccalaureate Questions

Questions for the Baccalaureate written papers are based on the physics syllabus of the 7th year, although they may call on knowledge acquired earlier. Knowledge of material covered only in the optional sections of the syllabus will not be assumed.

Questions should have a good balance of elements; not too much which is just recall, and not too much which demands very original thinking.

They should in the main examine the general understanding of physical principles, and not memory or the ability to substitute into formulae.

Questions should not be so long as to intimidate the pupil with the task of reading them thoroughly and understanding the material. This is particularly noteworthy for pupils who will be working in a foreign language because of the difficulty of opening options in smaller language sections.

Candidates answer these 4 questions using a different page for each question. Answers must be supported by explanations. They must show the reasoning behind the results or solutions provided. If graphs are used to find a solution, they must be sketched as part of the answer. Unless indicated otherwise, full marks will not be awarded if a correct answer is not accompanied by supporting evidence or explanations of how the results or the solutions have been achieved. When the answer provided is not the correct one, still some marks can be awarded if it is shown that an appropriate method and/or a correct approach has been used.

The IT-tools allowed during the examinations are defined in a separate document approved by the JTC.

Oral Baccalaureate Examinations

- The oral exams may not be considered as a repetition of the written exams.
- The oral examination consists of a presentation which may be media-supported, followed by a discussion and an examination interview.
- The candidates dispose of 20 minutes preparation time and the oral examination may not last longer than 20 minutes.
- The tasks to be worked on in the oral examination are based on S7 and topics of S6 may be needed to cover the exam.
- Teachers are expected to provide, along with the examination questions, the solution and a grid with expected answers and the marks to be allocated.

For all assessment, the marking scale of the European schools shall be used, as described in “*Marking system of the European schools: Guidelines for use*” (Ref.: 2017-05-D-29-en-7).

5.1. Attainment Descriptors

	A (9,0 - 10 Excellent)	B (8,0 - 8,9 Very good)	C (7,0 - 7,9 Good)	D (6,0 - 6,9 Satisfactory)	E (5,0 - 5,9 Sufficient)	F (3,0 - 4,9 Weak/Failed)	FX (0 - 2,9 Very weak/Failed)
Knowledge	Displays comprehensive knowledge of facts	Displays a very broad knowledge of facts	Displays a broad knowledge of facts	Displays a reasonable knowledge of facts and definitions	Recalls main names, facts and definitions.	Displays little recall of factual information	Displays very little recall of factual information.
Comprehension	and a thorough command and use of concepts and principles in science.	and a good command and use of concepts and principles in science.	and good understanding of main concepts and principles in science.	and understanding of basic concepts and principles in science.	Understands only basic concepts and principles in science	and a limited understanding of concepts and principles in science.	Shows very little understanding of scientific principles and concepts.
Application	Makes connections between different parts of the syllabus and applies concepts to a wide variety of unfamiliar situations and makes appropriate predictions.	Makes some connections between different parts of the syllabus and applies concepts and principles to unfamiliar situations.	Is capable of using knowledge in an unfamiliar situation.	Is capable of using knowledge in a familiar situation.	and can use basic knowledge in a familiar situation.	/	/
Analysis	Is capable of detailed and critical analysis and explanations of complex data.	Analyses and explains complex data well.	Produces good analysis and explanations of simple data.	Produces basic analysis and explanations of simple data.	Given a structure can analyse and explain simple data.	Can use data only with significant guidance.	Fails to use data adequately.

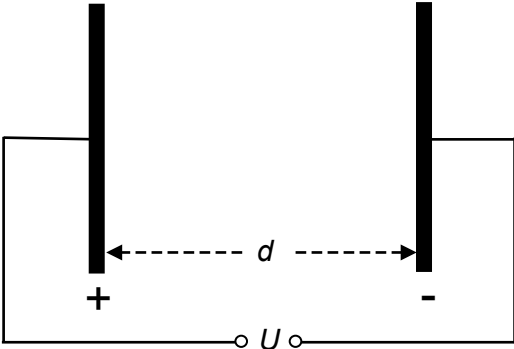
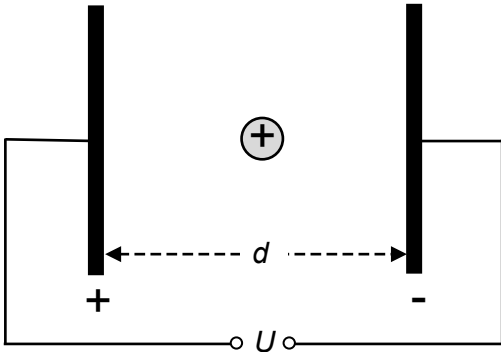
	A (9,0 - 10 Excellent)	B (8,0 - 8,9 Very good)	C (7,0 - 7,9 Good)	D (6,0 - 6,9 Satisfactory)	E (5,0 - 5,9 Sufficient)	F (3,0 - 4,9 Weak/Failed)	FX (0 - 2,9 Very weak/Failed)
Experimental work	Formulates hypotheses, plans and carries out investigations using a wide range of techniques while being aware of ethical issues.	Plans and carries out experiments using appropriate techniques, being aware of safety issues.	Follows a written procedure safely and makes and records observations, presenting them using different techniques.	Follows a written procedure safely and records observations.	Follows a written procedure safely and makes basic observations.	Has difficulty following instructions without supervision.	Is not able to safely follow a written procedure.
Digital and Information Competences*	Can consistently independently find and assess the reliability of, information on scientific subjects, on- and offline. Can independently use appropriate software for science tasks.	Can usually independently find and assess the reliability of, information on scientific subjects, on- and offline. Can use appropriate software for science tasks with some assistance.	Can often independently find and assess the reliability of, information on scientific subjects, on- and offline. Can use appropriate software for science tasks with assistance.	With aid, can find and assess the reliability of, information on scientific subjects, on- and offline. Can use appropriate software for science tasks given structured assistance.	Can retrieve information on scientific subjects when directed to reliable sources, on- and offline. Can follow structured instructions to use appropriate software for science tasks.	Generally unable to find, or to assess the reliability of, information on scientific subjects, on- and offline. Has great difficulties using appropriate software for science tasks even with assistance.	Unable to find, or to assess the reliability of, information on scientific subjects, on- or offline. Unable to use appropriate software for science tasks even with assistance.

	A (9,0 - 10 Excellent)	B (8,0 - 8,9 Very good)	C (7,0 - 7,9 Good)	D (6,0 - 6,9 Satisfactory)	E (5,0 - 5,9 Sufficient)	F (3,0 - 4,9 Weak/Failed)	FX (0 - 2,9 Very weak/Failed)
Communication (oral and written)	Communicates logically and concisely using scientific vocabulary correctly. Demonstrates excellent presentation skills.	Communicates clearly using scientific vocabulary correctly. Demonstrates very good presentation skills.	Communicates clearly most of the time using scientific vocabulary correctly. Demonstrates good presentation skills.	Uses basic scientific vocabulary, and descriptions show some structure. Demonstrates satisfactory presentation skills.	Uses basic scientific vocabulary, but descriptions may lack structure or clarity. Demonstrates satisfactory presentation skills.	Generally produces descriptions that are insufficient or incomplete with a poor use of scientific vocabulary. Lacks acceptable presentation skills.	Has very poor Communication and presentation skills.
Teamwork	Shows initiative – a team leader.	Works constructively in a team.	Works well in a team.	Works satisfactorily in a team.	and participates in teamwork.	Needs assistance when working in a team.	Does not work in a team.

*) This competence is part of the European Digital Competence Framework (<https://ec.europa.eu/jrc/en/digcomp>)

6. Annex

6.1. Example Baccalaureate Exam

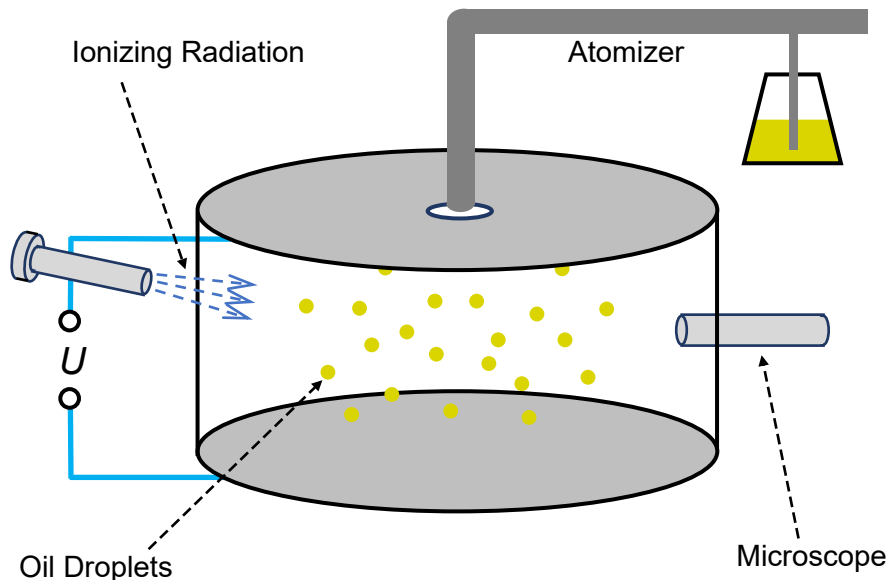
Question 1: Fields		
	Page 1/3	Marks
a) A capacitor is charged up to a potential difference U between the plates.		
		
The electric field between charged bodies is characterized by the field lines.		
i. Describe three of the principles according to which the electric field lines are drawn between charged conductive surfaces.		3
ii. Copy the two plates of the above figure and sketch the field lines and equipotentials.		2
iii. Now a small positively charged ball is put in the middle between the two plates.		
		
Copy the two plates and the ball, and sketch the field lines.		2

Question 1: Fields

Page 2/3

Marks

- b)** In his historic experiment Robert Millikan determined the elementary electric charge by spraying oil droplets into a vertically directed electric field of a capacitor.



The drop can be observed via a microscope attached to the apparatus, and dependent on the potential difference applied to the plates of the capacitor it moves upward, downward, or does not move.

- i. Describe how this experiment is conducted, paying particular attention to the following points:
 - the purpose of the X-rays;
 - the purpose of the microscope;
 - the conditions for a droplet to move upward, downward, or float.
- ii. The radius r of the droplet can be measured and allows to calculate its mass m .

Neglecting buoyancy, the charge q of a single floating oil droplet can then be determined with the equation

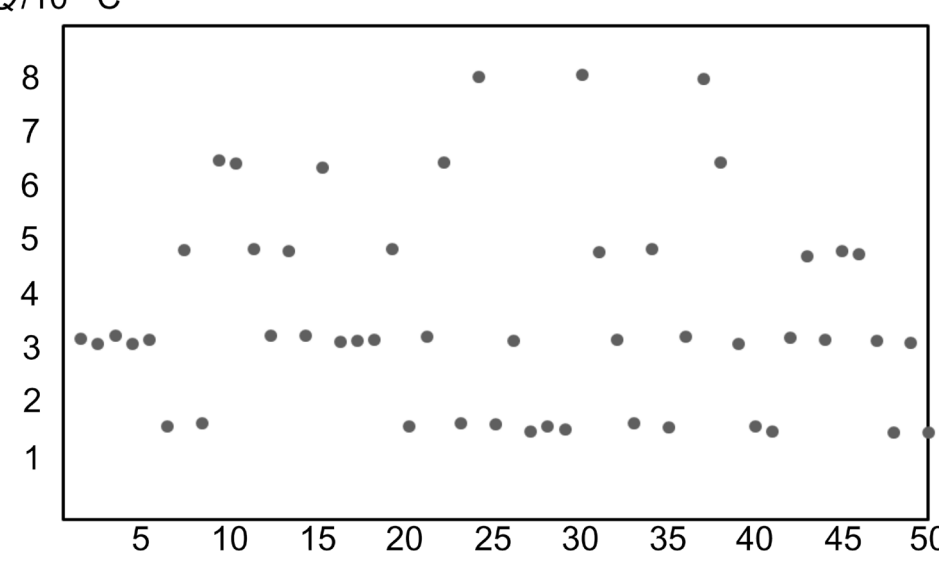
$$q = \frac{m g d}{U},$$

where g the acceleration due to gravity, d the plate distance and U the applied electric voltage.

Derive this equation.

4

3

Question 1: Fields																							
	Page 3/3	Marks																					
iii. The figure below shows the results of a measurement series of 50 oil droplets. $Q/10^{-19}\text{C}$  Interpret these results. c) In a second series of experiments, the measured values for 6 oil droplets shown were recorded to determine the elementary charge at a plate distance $d = 2.76\text{ mm}$. <table border="1" data-bbox="234 1274 1137 1449"><tr><td>experiment</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>U/V</td><td>52</td><td>174</td><td>88</td><td>117</td><td>138</td><td>92</td></tr><tr><td>$m/10^{-16}\text{ kg}$</td><td>6.16</td><td>10.2</td><td>15.4</td><td>7.05</td><td>16.7</td><td>22.0</td></tr></table> Using the formula given in part b)ii (with $g = 9.81\text{ ms}^{-1}$), cError! Bookmark not defined.alculate the charge q of each oil droplet, and explain (without calculation) how to find the elementary charge. d) Explain the significance of the Millikan experiment for the development of quantum physics.			experiment	1	2	3	4	5	6	U/V	52	174	88	117	138	92	$m/10^{-16}\text{ kg}$	6.16	10.2	15.4	7.05	16.7	22.0
experiment	1	2	3	4	5	6																	
U/V	52	174	88	117	138	92																	
$m/10^{-16}\text{ kg}$	6.16	10.2	15.4	7.05	16.7	22.0																	
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		2																					

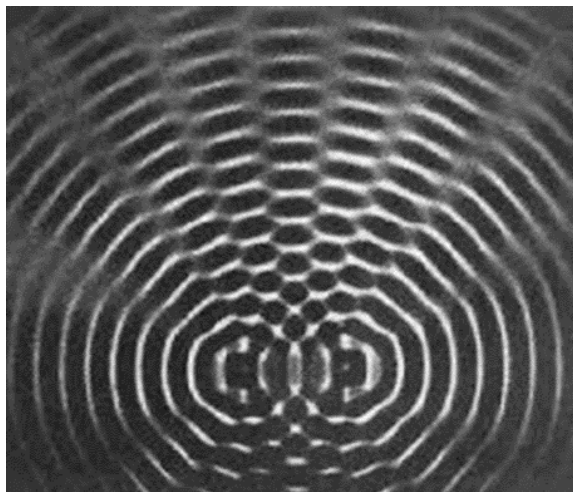
Question 2: Oscillations and Waves

Page 1/2

Marks

- a) i.** In a ripple tank, waves are generated by two coherent sources oscillating in phase. This creates an interference pattern on the water surface.

Explain the origin of this interference pattern.



4

- ii.** Far away from the sources the interference maxima are located (to a good approximation) on straight lines that run through the middle of the connecting line of the sources. For the angle α between these straight lines and the line segment bisector (equidistant line) to the two sources, the following formula applies,

$$\sin \alpha_k = \frac{k \lambda_k}{d}$$

where d is the distance between the sources.

Derive this formula.

2

- iii.** The distance between two ultrasound generators is $d = 20.0$ mm. The angle between the directions to distant maxima of 0th and 1st order is 12° .

(The ultrasound range starts at around 20 kHz.)

Show that the frequency of the ultrasound is about $f = 82$ kHz.

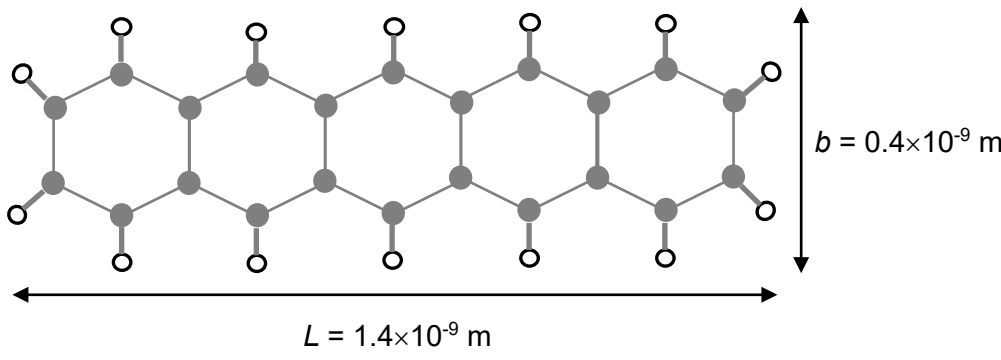
2

- iv.** Justify, using sketches similar to the ripple tank picture above, if changing the distance d has an impact on the number of maxima observed.

3

Question 2: Oscillations and Waves

	Page 2/2	Marks
b) Bats regularly emit short sound signals in the ultrasonic range and orient themselves by the returning echo of objects or prey. The nostrils of one species of bats (the Greater Horseshoe Nose) are only about 2 mm apart. It uses locating sounds of $f = 82 \text{ kHz}$, which it emits through its nostrils. The speed of sound in air is 340 m s^{-1}. Source (2020-12-23): https://unsplash.com/photos/_f8lZ0gGS6E 		
i. Verify that only the 0th-order maximum of interference will exist.		4
ii. Describe a disadvantage for echolocation if there were higher order maxima.		3
c) i. The intensity of the echo is sufficient to be recognized by the bat up to a distance of about 30 m. The bat is flying directly towards the prey with a speed of $u = 15 \text{ km/h}$. Calculate the time it takes the sound to travel out and back, taking into consideration the movement of the bat.		4
ii. Ships also use echo location to detect obstacles like rocks or other ships, but they use electromagnetic waves in the microwave range (also called radar waves) instead of sound waves.		
1) Give the range of the wavelengths for microwaves.		1
2) Calculate the time it takes for radar waves to move out and back between the ship and an obstacle 3.0 km away. Explain why, the speed of the ship (less than 50 km/h) needs not be taken into consideration here.		2

Question 3: Quantum Physics		
	Page 1/2	Marks
a) The potential well with infinitely high walls helps to model the quantisation of the energy levels of a particle "confined" in a space without too much mathematical effort. The energy levels for an infinite potential well with width L are as follows: $E_n = \frac{h^2}{8 m_e L^2} n^2 \quad n = 1, 2, 3, \dots$		
i. Describe the model of the infinite potential well and explain how this model explains the quantization of the energy.		3
ii. Draw the matter waves for the three lowest levels.		3
b) The model can be applied to the molecule Pentacene.		
		
i. Using the uncertainty principle, explain why the lowest energy state for the Pentacene molecule has an energy greater than zero.		2
ii. For the molecule Pentacene, calculate the ground-state energy in eV due to the length L of this molecule.		2

Question 3: Quantum Physics		
	Page 2/2	Marks
iii. Determine whether the wavelength of the photon emitted at the transition of an electron from state 4 to state 2 is in the range of visible light. iv. In a quantum system the total energy is given by the sum of the energies due to the individual quantum numbers. Why does the width b of the molecule not affect the result in question b) iii.?		5
c) Even if atoms do not have an infinite potential well, the physical principles learned from the model discussed above also applies to them. For the hydrogen atom, the energy levels are given by $E_n = -13.6 \text{ eV} \times \frac{1}{n^2} \quad n = 1, 2, 3, \dots$		2
i. Explain what is meant by ionisation energy.		2
ii. Show that the energy necessary to ionize a hydrogen atom which is in its ground state, is 13.6 eV.		1
iii. Calculate the minimum momentum of a photon that can ionise this hydrogen atom.		3
iv. Explain whether there can be ionisation in the infinite well.		2

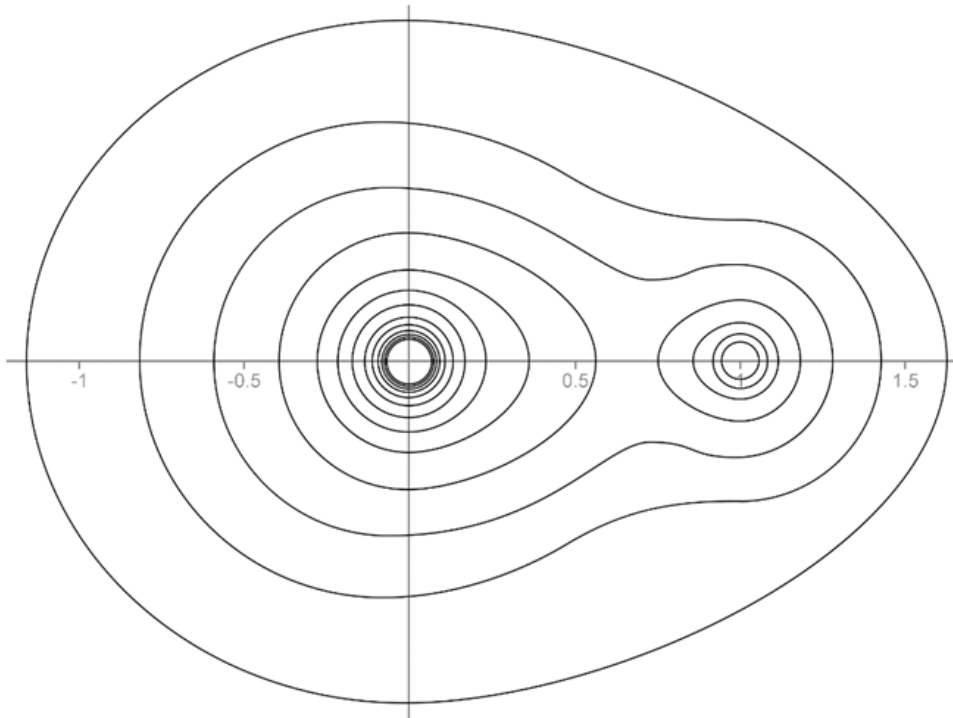
Data:	
Planck constant:	$h = 6.63 \times 10^{-34} \text{ J s}$
mass of the electron:	$m_e = 9.11 \times 10^{-31} \text{ kg}$
speed of light in vacuum:	$3.00 \times 10^8 \text{ m s}^{-1}$
elementary charge:	$e = 1.60 \times 10^{-19} \text{ C}$
range of visible light:	$380 \text{ nm} \leq \lambda \leq 740 \text{ nm}$

Question 4: Mixed Topics

Page 1/2

Marks

a)



On the figure above, you can see a few equipotentials of the Earth-Moon system at a given time.

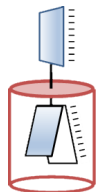
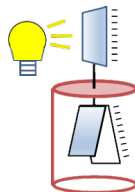
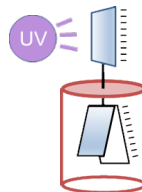
- i. Find the (x,y) coordinates of the Earth and of the Moon on the figure above. Justify. 2
- ii. Use the figure above to sketch the potential as a function of distance from the Earth, on the line segment between the Earth and the Moon. On your sketch, identify the location of the point in space where the gravitational forces of the Earth and the Moon cancel each other (the so-called null point). 4

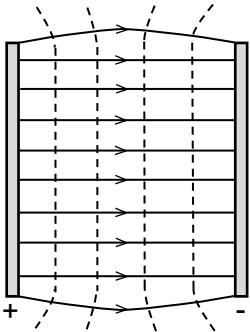
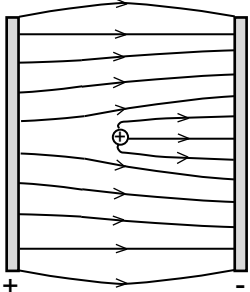
- b) In order to get the acceleration on the surface of the Moon, an astronaut measures the period of oscillation T of a simple pendulum of variable length L on the Moon. He obtains the following results:

L / m	0.200	0.300	0.400	0.500	0.600
T / s	2.1	2.8	3.0	3.5	3.8

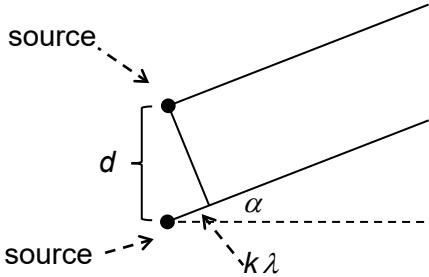
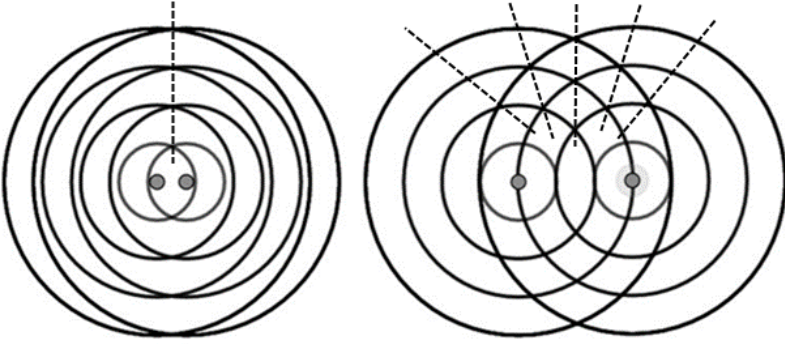
- i. Draw a graph giving L as a function of T^2 3
- ii. Explain how to obtain g_{Moon} from these values, knowing that 3

$$T = 2\pi \times \sqrt{\frac{L}{g_{\text{Moon}}}}$$

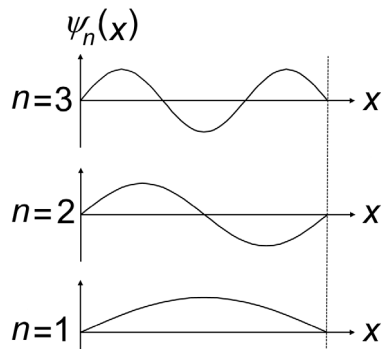
Question 4: Mixed Topics				
			Page 2/2	Marks
c)	Charged zinc plate Electroscope 	α) Lightbulb 	β) UV 	
	In the picture above, two important experiments are performed. A zinc plate is charged at the beginning of both the α and β experiment. In the α experiment a regular incandescent lightbulb is used, and the zinc plate stays charged. In the β experiment a UV lamp is used, and the zinc plate discharges. i. This must have been disturbing for physicists in the 1900s. Explain why. ii. When the β experiment is repeated but a pane of glass is inserted between the UV lamp and the charged zinc plate, it stays charged. Use this to explain how one might feel hot in a glasshouse but should not get a sunburn.			
d)	When a wave (mechanical, electromagnetic or quantum) passes through an opening, diffraction occurs. The effect is negligible if $\lambda \ll d$. Explain how you do not appear to diffract when passing through a door.			3
e)	i. The physicist Bohr, in his model of the Hydrogen atom, suggested that electrons move around the proton on a circular orbit. In classical physics, an accelerated charge will emit electromagnetic radiation. Following classical mechanics, describe the fate of the electron in a Hydrogen atom. ii. The hydrogen atom is made of one electron and one proton, feeling mutual electrostatic attraction. A classical lowest energy configuration would then have the electron directly on top of the proton. Find a quantum physics principle which forbids the electron to reach the proton.			3 2

Solution to Question 1					Fields				
A: Knowledge and Comprehension; B: Application; C: Analysis and Evaluation; W: Written Communication					A	B	C	W	Σ
a)	i.	3 Principles should be mentioned, e.g.: • The field lines of an electric field end perpendicular to conductive surfaces. • Field lines do not cross each other. • Electric Field lines are pointing from positive to negative charges. • Field lines are drawn at shortest way between charged surfaces, but at largest possible distance to neighbouring lines. • ...			3				3
	ii.				2				2
	iii.	 Any solution where at least one additional field line goes from the additional charge to the negative plate, and where some lines going from the positive to the negative plate are curbed away from the charge, should be accepted.				1	1		2
b)	i.	Oil droplets are sprayed into the container. A droplet which sinks into the capacitor is irradiated by the x-rays and becomes electrically charged. The microscope allows precise observation of the motion of the droplets. If the gravitational force exceeds the electric force, the droplet moves downwards, else it moves upwards. If both forces cancel each other, the droplet floats. This situation can be achieved by adjusting the potential difference between the plates of the capacitor.			1.5			1	4
	ii.	For a floating droplet we have $F_{\text{el}} = F_g \Rightarrow E q = \frac{U}{d} q = m g \Rightarrow q = \frac{m g d}{U}$			1.5				
	ii.	For a floating droplet we have $F_{\text{el}} = F_g \Rightarrow E q = \frac{U}{d} q = m g \Rightarrow q = \frac{m g d}{U}$			1	2			3

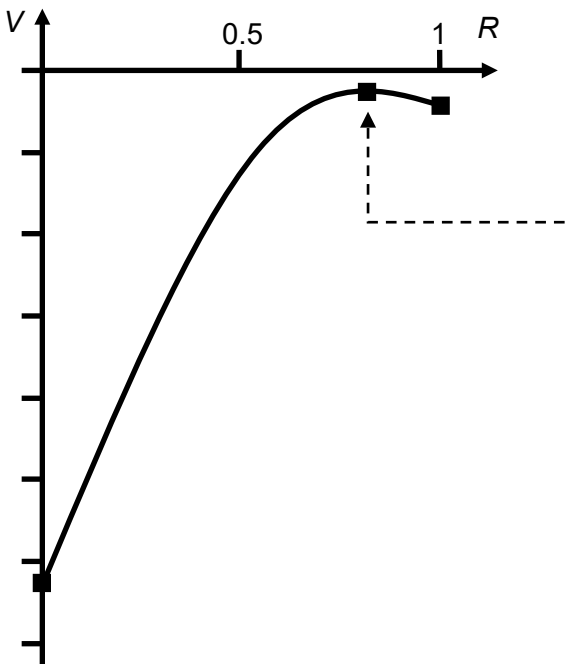
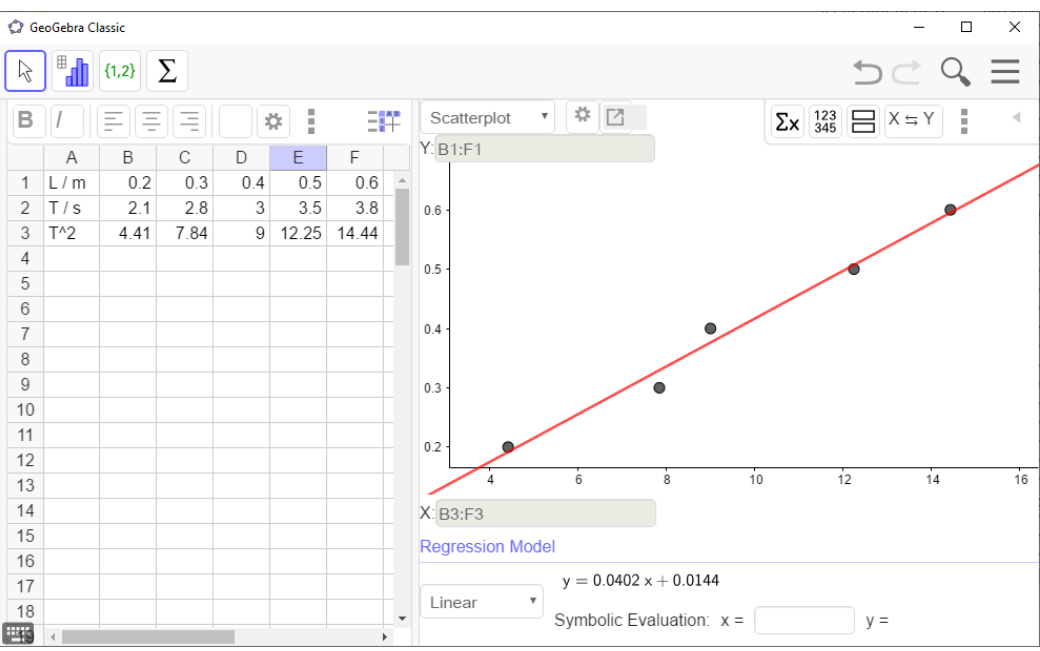
Solution to Question 1							Fields																																																																		
iii.	The points lie approximately along several horizontal lines. It can be seen that the oil droplets only have charges q whose values are an integer multiple of a certain charge e : $q = n e$ and $n = \text{integer}$. This particular charge e is the elementary charge. The deviations from the values $q = n e$ can be attributed to the inaccuracies of the measurement.						2	1	1	4																																																															
c)	With Geogebra we get for the charges: <table><tr><td>droplet</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>U / V</td><td>52</td><td>174</td><td>88</td><td>117</td><td>138</td><td>92</td></tr><tr><td>m/10⁻¹⁶kg</td><td>6.16</td><td>10.2</td><td>15.4</td><td>7.05</td><td>16.7</td><td>22</td></tr><tr><td>q/10⁻¹⁹C</td><td>3.21</td><td>1.59</td><td>4.74</td><td>1.63</td><td>3.28</td><td>6.47</td></tr></table> Find the lowest value of q in the above table. Then divide all charges by this minimum charge to check whether they are multiples of this value. If yes (within the margin of error), it is the elementary charge. The calculation is not required. Nevertheless, it is shown here: The lowest charges are no. 2 and 4. (So its best to use their average.) <table><tr><td>droplet</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>U / V</td><td>52</td><td>174</td><td>88</td><td>117</td><td>138</td><td>92</td></tr><tr><td>m/10⁻¹⁶kg</td><td>6.16</td><td>10.2</td><td>15.4</td><td>7.05</td><td>16.7</td><td>22</td></tr><tr><td>q/10⁻¹⁹C</td><td>3.21</td><td>1.59</td><td>4.74</td><td>1.63</td><td>3.28</td><td>6.47</td></tr><tr><td>q / q_{2.4}</td><td>1.99</td><td>0.99</td><td>2.94</td><td>1.01</td><td>2.04</td><td>4.02</td></tr></table> Within the margin of error all charges are integer multiples.						droplet	1	2	3	4	5	6	U / V	52	174	88	117	138	92	m/10 ⁻¹⁶ kg	6.16	10.2	15.4	7.05	16.7	22	q/10 ⁻¹⁹ C	3.21	1.59	4.74	1.63	3.28	6.47	droplet	1	2	3	4	5	6	U / V	52	174	88	117	138	92	m/10 ⁻¹⁶ kg	6.16	10.2	15.4	7.05	16.7	22	q/10 ⁻¹⁹ C	3.21	1.59	4.74	1.63	3.28	6.47	q / q _{2.4}	1.99	0.99	2.94	1.01	2.04	4.02	2.5			5
droplet	1	2	3	4	5	6																																																																			
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d)	The important difference between classical and quantum physics is quantisation. The quantisation of the electric charge was the first being discovered thanks to the Millikan experiment. That helped to get the idea that other physical values might also be quantised.							1	1	2																																																															
Sum:							9	9	4	3	25																																																														

Solution to Question 2			Fields				
A: Knowledge and Comprehension; B: Application; C: Analysis and Evaluation; W: Written Communication			A	B	C	W	Σ
a)	i.	Depending on the path difference of the waves emitted by the two sources, constructive or destructive interference takes place. Constructive interference occurs with a path difference of $k\lambda$, destructive interference occurs for an additional half wavelength.	2			2	4
	ii.	 For constructive interference to occur at a point the path difference between the waves meeting at that point must be equal to a whole number multiple of the wavelength of the light. $\Rightarrow \sin \alpha = \frac{k \lambda}{d}$		2			2
	iii.	$\sin \alpha_n = \frac{k \lambda}{d} \Rightarrow \lambda = d \sin \alpha_1 = 20 \text{ mm} \times \sin 12^\circ = 0.02 \text{ m} \times \sin 12^\circ$ $= 0.00416 \text{ m} = 4.16 \text{ mm}$ $c = \lambda f \Rightarrow f = \frac{c}{\lambda} = \frac{340 \text{ m s}^{-1}}{0.00416 \text{ m}} = 81.7 \text{ kHz}$	1				2
	iv.	Only the crest of the waves need to be shown, as usually done applying the Huygens principle. The sketches can be produced with Geogebra.  The greater the distance, the more maxima you obtain.		2		1	3
b)	i.	The longest possible path difference appears for waves going along the line connection the two sources. Therefore, the path difference can at most become as large as d. If the wavelength is greater than d, only the maximum of 0th order can occur. The wavelength is $f \lambda = v \Rightarrow \lambda = \frac{v}{f} = \frac{340 \text{ m s}^{-1}}{82 \times 10^3 \text{ s}^{-1}} = 0.0041 \text{ m} = 4.1 \text{ mm} > 2 \text{ mm}$ For $f = 82 \text{ kHz}$ the wavelength is greater than the distance of the bat's nostrils. So only the 0th order maximum can occur.	1	1			4
					2		

Solution to Question 2				Fields				
	ii.	It is sufficient if one of the following possibilities or another plausible explanation is given: - Straight ahead the intensity is particularly high, towards the side it decreases; - With secondary maxima there would also be maxima from which no echo would come back; - The localization is made easier because there are no higher order maxima. - ...			1	1	1	3
c)	i.	Sound: $s_{\text{sound}} = v t$ Bat: $s_{\text{bat}} = u t$ $2 \times 30\text{m} - s_{\text{bat}} = s_{\text{sound}}$ $2 \times 30\text{m} = s_{\text{sound}} + s_{\text{bat}} = v t + u t = (v + u) t$ $\Rightarrow t = \frac{2 \times 30\text{m}}{v + u} = \frac{2 \times 30\text{m}}{340\text{ms}^{-1} + 15 \times 10^3 \text{m}/(3600\text{s})^{-1}} = 0.17\text{s}$		2	1	1		4
	ii.	1)	From the formula booklet: $10^{-1}\text{m} > \lambda > 10^{-3}\text{m}$	1				1
		2)	$s = c t \Rightarrow t = \frac{s}{c} = 2 \frac{3 \times 10^3\text{m}}{3 \times 10^8 \frac{\text{m}}{\text{s}}} = 2 \times 10^{-5}\text{s} (= 20 \mu\text{s})$ The speed of the ship is much smaller than the speed of light. In part i, the relation is $\frac{340\text{ms}^{-1}}{15 \times 10^3 \text{m}/(3600\text{s})^{-1}} = 82.$ Here it is $\frac{3 \times 10^8 \text{ms}^{-1}}{50 \times 10^3 \text{m}/(3600\text{s})^{-1}} \approx 7 \times 10^7$	1	1			2
Sum:				8	9	4	4	25

Solution to Question 3			Fields				
A: Knowledge and Comprehension; B: Application; C: Analysis and Evaluation; W: Written Communication			A	B	C	W	Σ
a)	i.	- The potential energy of the electron inside the potential well is zero. At the edge of the well it becomes infinite. The matter wave representing the electron is completely reflected at this wall, and a standing wave is formed between the two sides of the well, like on a string fixed at both sides. - This standing wave can only exist if an integer multiple of half the wavelength fits into the width of the well. The quantization of the energy is a consequence of the dependence of the energy on the wavelengths of the standing waves.	0.5			0.5	3
	ii.		3				3
b)	i.	The uncertainty principle states that position and momentum cannot simultaneously be known with perfect precision. Since the molecule Pentacene has a finite size, the momentum of the electron - and its energy - cannot be known exactly and thus cannot be zero.		1	0.5	0.5	2
	ii.	For the lowest state $n = 1$, the formula gives $E_n = \frac{h^2}{8 m_e L^2} n^2$ $E_1 = \frac{(6.63 \times 10^{-34} \text{ J s})^2}{(8 \times 9.11 \times 10^{-31} \text{ kg})(1.4 \times 10^{-9} \text{ m})^2} \times 1^2 = 3.01 \times 10^{-20} \text{ J} = 0.19 \text{ eV}$		2			2
	iii.	$\Delta E = E_4 - E_2 = \frac{(6.63 \times 10^{-34} \text{ J s})^2}{(8 \times 9.11 \times 10^{-31} \text{ kg})(1.4 \times 10^{-9} \text{ m})^2} (4^2 - 2^2) = 3.69 \times 10^{-19} \text{ J}$ The corresponding wavelength of the emitted light is $E = hf \Rightarrow \lambda = \frac{hc}{E} = \frac{(6.63 \times 10^{-34} \text{ J s})(3 \times 10^8 \text{ m s}^{-1})}{3.69 \times 10^{-19} \text{ J}} = 5.39 \times 10^{-7} \text{ m} = 539 \text{ nm}$ This is in the range of visible light.		5			5

Solution to Question 3		Fields				
	iv. - A change of the quantum number due to b was not considered in b) ii. - As the energies connected with this quantum number are much higher, corresponding photons are outside the range of visible light. - ... (Giving one reason is sufficient for full marks.)			1.5	0.5	2
c)	i. That is the energy to excite an electron to an infinitely high level and thus free it from the atom.	1.5			0.5	2
	ii. The energy in the ground state $n = 1$ is 13.6 eV, for $n = \infty$ it is zero. The ionisation energy therefore is 13.6 eV.	1				1
	iii. The energy of the photon must be greater or at least equal to the ionisation energy. $p = \frac{h}{\lambda} = \frac{h f}{c}; \quad E = h f \Rightarrow f = \frac{E}{h}$ $\Rightarrow p = \frac{h E}{c h} = \frac{E}{c} = \frac{13.6 \text{ eV}}{3 \times 10^8 \text{ m s}^{-1}} = \frac{13.6 \text{ eV} \times 1.6 \times 10^{-19} \frac{\text{J}}{\text{eV}}}{3 \times 10^8 \text{ m s}^{-1}} = 7.25 \times 10^{-27} \text{ kg m s}^{-1}$	1	1			3
	iv. As the well has infinite height, the electron cannot leave it.			1.5	0.5	2
Sum:		9.5	9	3.5	3	25

Solution to Question 4:			Fields				
A: Knowledge and Comprehension; B: Application; C: Analysis and Evaluation; W: Written Communication			A	B	C	W	Σ
a)	i.	Earth is at (0,0), while Moon is at (1,0). The masses generating the fields must be at the centre of the (circular) equipotentials. Because of the Earths stronger field, its equipotentials must be closer together than those of the Moon.	2				2
	ii.	 The null point is located where the potential graph is horizontal, because a mass moves toward a lower potential.		2.5	1	0.5	4
b)	i.	 The pupil should know (or see from the formula given) that for a pendulum $L \propto T^2$, and thus the graph should be a straight line.	2	1			3

Solution to Question 4:			Fields			
	ii	Alternative 1: via the graph $T = 2\pi \sqrt{\frac{L}{g_{\text{Moon}}}} \Rightarrow L = \frac{g_{\text{Moon}}}{4\pi^2} T^2$ The points must be located on a straight line of slope $s = \frac{g_{\text{Moon}}}{4\pi^2}$. Alternative 2: from the table $T = 2\pi \sqrt{\frac{L}{g_{\text{Moon}}}} \Rightarrow g_{\text{Moon}} = L \frac{4\pi^2}{T^2}$ For each table column, you have to calculate g. Then, to increase precision, you take the average of these results.	2.5		0.5	3
c)	i.	In classical wave mechanics, all waves carry energy in a continuous manner. For a classical physicist, then the only difference between the incandescent and UV lamp is the power transmitted. They would thus expect either both lamps to discharge, or both lamps to fail at discharging.	2		1	3
	ii.	The new experiment tends to show that UV light is blocked by glass. One can expect visible light to go through glass. A glasshouse relies on visible light going into the glasshouse, being converted to IR, and thus raising the temperature. UV light being blocked by glass, there shouldn't be cause for sunburns in the glasshouse.	1	0.5	0.5	2
d)		Students should discuss the wavelength for the matter wave associated with macroscopic objects. The bigger the momentum of an object, the shorter wavelength of the corresponding matter wave. For a macroscopic object, the wavelength is extremely short (especially compared to the width of a door). Thus diffraction should not occur.	1	1	1	3
e)	i.	Circular motion is an accelerated motion, and thus orbiting electrons should radiate energy away in a continuous manner. The total mechanical energy of an electron on this circular orbit depends on the radius as well, with smaller radii corresponding to smaller energies. Thus the electron should orbit on ever smaller radius; the trajectory will be a spiral.	1	1	1	3
	ii.	Heisenberg's uncertainty principle states that measurements cannot be performed with an infinite precision on both position and momentum of a quantum particle. The energy of the electron is in the range of the atom's energy levels. If the electron were at rest on the proton, it would be spread out too much. (Alternative: The electron being at an exact place on top of the proton would lead to a large momentum which would move it away.)	1	1	1	2
Sum:			8	9	3.5	25

6.2. Generic Matrix for Physics

Element of examination	Competence	Weight in %	Evaluation/Marking	Weight in marks
Fields	Knowledge and Comprehension	$\pm 36 \%$	Paper-specific Marking Scheme	± 9
	Application	$\pm 36 \%$		± 9
	Analysis and Evaluation	$\pm 16 \%$		± 4
	Written Communication	$\pm 12 \%$		± 3
		100 %		25
Oscillation and Waves	Knowledge and Comprehension	$\pm 36 \%$	Paper-specific Marking Scheme	± 9
	Application	$\pm 36 \%$		± 9
	Analysis and Evaluation	$\pm 16 \%$		± 4
	Written Communication	$\pm 12 \%$		± 3
		100 %		25
Quantum Physics	Knowledge and Comprehension	$\pm 36 \%$	Paper-specific Marking Scheme	± 9
	Application	$\pm 36 \%$		± 9
	Analysis and Evaluation	$\pm 16 \%$		± 4
	Written Communication	$\pm 12 \%$		± 3
		100 %		25
Mixed Topics	Knowledge and Comprehension	$\pm 36 \%$	Paper-specific Marking Scheme	± 9
	Application	$\pm 36 \%$		± 9
	Analysis and Evaluation	$\pm 16 \%$		± 4
	Written Communication	$\pm 12 \%$		± 3
		100 %		25
Total Exam				100

In each section a deviation of up to 6% (1.5 points) will be tolerated as long as the total number of marks (25 points) is respected for each question.

In the EB, a student will be deemed to have reached sufficient at 50 points out of 100.

EXCELLENT	VERY GOOD	GOOD	SATISFACTORY	SUFFICIENT	WEAK	VERY WEAK
100-90	89-80	79-70	69-60	59-50	49-21	20-0

6.3. Paper-specific Matrix for the Example Exam

Element of examination	Competence	Weight in %	Evaluation/Marking	Weight in marks
Fields	Knowledge and Comprehension	36.0 %	Paper-specific Marking Scheme see solutions	9.0
	Application	36.0 %		9.0
	Analysis and Evaluation	16.0 %		4.0
	Written Communication	12.0 %		3.0
		100 %		25
Oscillation and Waves	Knowledge and Comprehension	32.0 %	Paper-specific Marking Scheme see solutions	8.0
	Application	36.0 %		9.0
	Analysis and Evaluation	16.0 %		4.0
	Written Communication	16.0 %		4.0
		100 %		25
Quantum Physics	Knowledge and Comprehension	38.0 %	Paper-specific Marking Scheme see solutions	9.5
	Application	36.0 %		9.0
	Analysis and Evaluation	14.0 %		3.5
	Written Communication	12.0 %		3.0
		100 %		25
Mixed Topics	Knowledge and Comprehension	32.0 %	Paper-specific Marking Scheme see solutions	8.0
	Application	36.0 %		9.0
	Analysis and Evaluation	14.0 %		3.5
	Written Communication	18.0 %		4.5
		100 %		25
Total Exam				100

6.4. Oral Record Sheet

Assessment Record Sheet Oral Examination – Physics

European School:

Student:

Class:

Examiner:

Mark	Evaluation according to the attainment descriptors
9.0-10 Excellent	The candidate can easily and unaided identify the topic of chosen questions and shows ability to set the given problem in a physics context, applies techniques of analysis with no aid, presents clear communication skills, uses appropriate vocabulary, recalls and applies the necessary definitions and laws, shows a coherent approach, demonstrates mastery of any computational techniques while presenting and making calculations, is at ease analysing unfamiliar problems, and can propose imaginative experimental or theoretical approaches, has clearly and concisely organized responses to set problems and engages fluently in follow-up discussion.
8.0-8.9 Very good	The candidate can identify the topic of chosen questions and shows ability to set the given problem in a physics context, applies techniques of analysis with little aid, presents clear communication skills, uses appropriate vocabulary, recalls and applies the necessary definitions and laws, shows a coherent approach, demonstrates proper computational techniques while presenting and making calculations, is able to analyse unfamiliar problems competently and can propose some investigative approaches, has well-organized responses to set problems, and is able to respond appropriately to follow-up discussion.
7.0-7.9 Good	The candidate can identify the topic of chosen questions and shows ability to set the given problem in a physics context with help from the examiners, is able to apply the techniques of analysis with some help, uses correct vocabulary, recalls and applies necessary definitions and laws with little aid, demonstrates proper computational techniques while presenting and making calculations, can analyse unfamiliar problems and suggest investigative approaches given little aid, has competently organized responses to set problems, and is able to engage in follow-up discussion, supported by the examiners.
6.0-6.9 Satisfactory	The candidate demonstrates understanding of major areas covered by chosen questions and shows ability to set the given problem in a physics context with some help from the examiners, uses correct vocabulary, recalls and applies some definitions and laws with aid, may require help to analyse unfamiliar problems or to suggest investigative approaches, has prepared and structured responses to set problems, and is able to engage in follow-up discussion when led by the examiners.

5.0-5.9 Sufficient The candidate demonstrates understanding of major areas covered by chosen questions and shows ability to set the given problem in a physics context with aid from the examiners, largely employs correct vocabulary and is able to recall basic facts related to questions, requires help to analyse unfamiliar problems, has prepared only few responses to set problems, and is able to engage in follow-up discussion with aid from the examiners.

3.0-4.9 Failed / Weak The candidate displays gaps in understanding of major areas covered by chosen questions, is unable to show ability to set the given problem in a physics context even with aid from the examiners and shows gaps in vocabulary and basic facts related to questions. Responses are missing to the majority of chosen questions. The candidate presents minimal capacity for follow-up discussion.

0-2.9 Failed / Very Weak The candidate displays gaps in understanding of major areas covered by chosen questions, is unable to show ability to set the given problem in a physics context even with aid from the examiners and shows major gaps in vocabulary and factual knowledge related to questions. Responses are substantially missing or incorrect. There is no possibility of follow-up discussion.

Final mark:

Date: **Signature:**